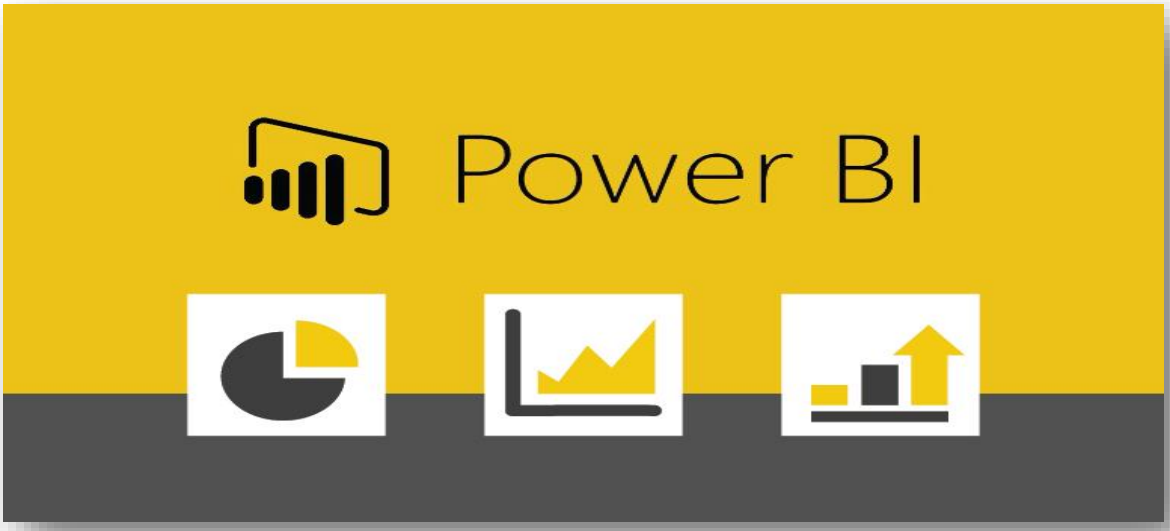
❖ POWER BI ASSIGNMENT 1 –E-COMMERCE DATA TRANSFORMATION & DATA MODELLING



🎯 **Project Objective:**

💼**Business Problem**:

- Analyse product order distributions, category profit margins, and sub-category sales performance to identify key growth drivers and optimize the store's product portfolio.

📌**Objective**:

- This project focuses on organizing and analysing different types of datasets.

The datasets are Sales Target, List of Orders, and Order Details.
Using Power BI, the workflow transforms raw datasets into a structured summary.

📁**Dataset Description:**

- This e-commerce dataset tracks product records across three primary categories: Clothing, Electronics, and Furniture.
- Each entry details product characteristics, including descriptive item names, specific sub-category classifications, and sales targets.
- For financial and operational reporting, the dataset maps the sales amount of individual items against their category and sub-category.
- It provides critical operational insights for dashboard construction and highlights category profit margins and sub-category sales performance to identify growth opportunities and optimize the store's product portfolio.

📋 Column Description

1. List of Orders Table

Column Name	Column Description
Order ID	Shows the order ID.
Order Date	Shows the order date.
Customer Name	Shows the customer name.
State	Shows the customer's state.
City	Shows the customer's city.

2. Order Details Table

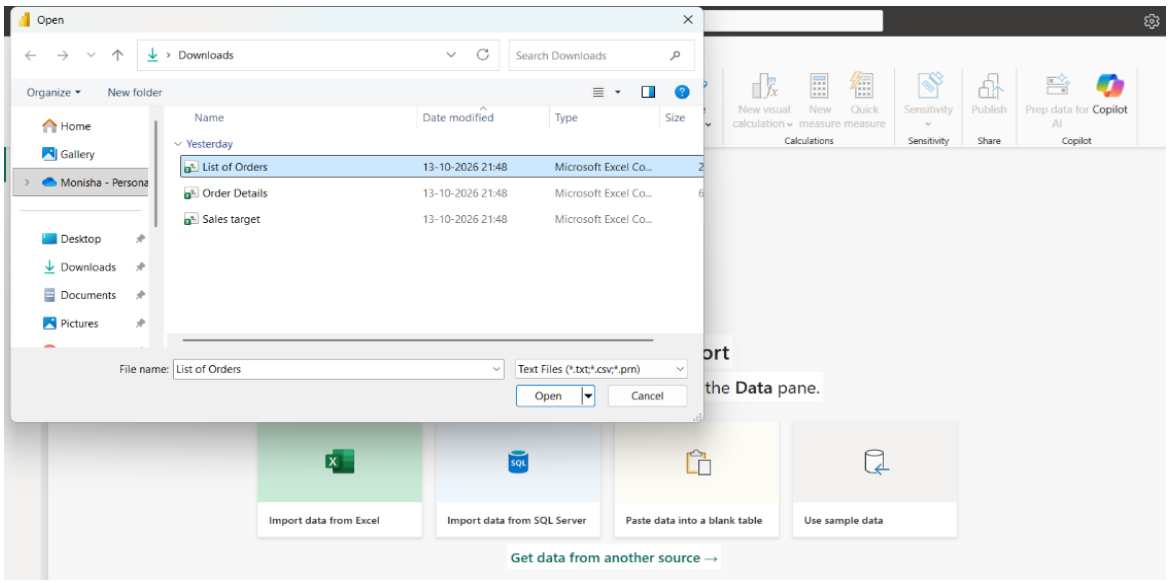
Column Name	Column Description
Order ID	Shows the order ID.
Amount	Shows the total amount of the product ordered by the customer.
Profit	Shows the company profit.
Quantity	Shows the quantity of products purchased by the customer.
Category	Shows the category of the product ordered by the customer.
Sub-Category	Shows the sub-category of the product ordered by the customer.

3. Sales Target Table

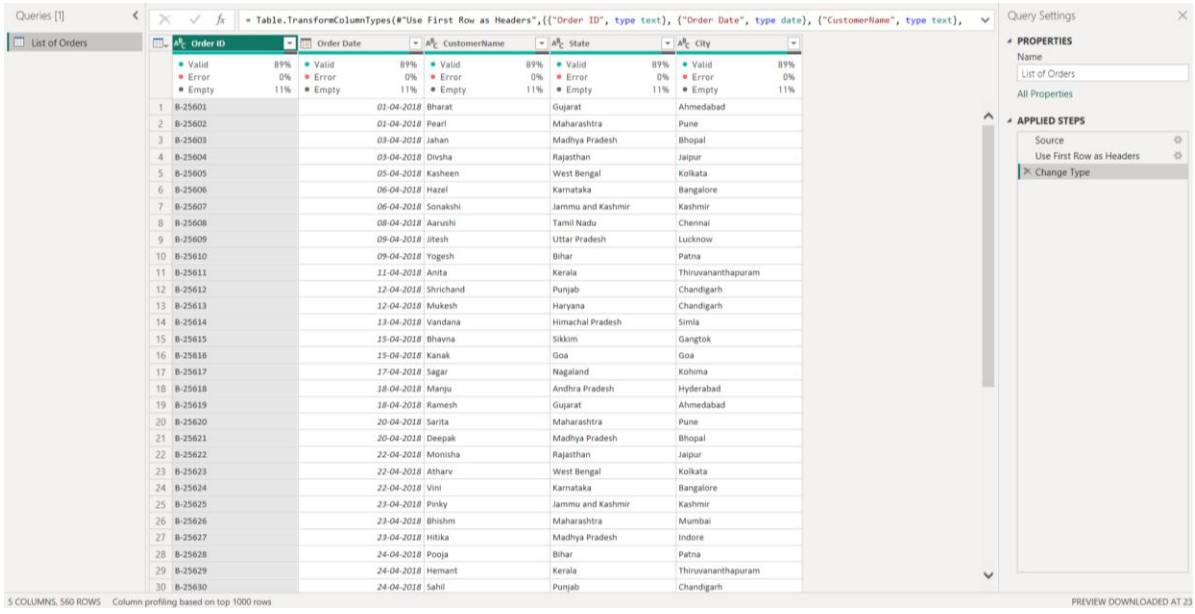
Column Name	Column Description
Month of Order Date	Shows the month in which the product was ordered.
Category	Shows the category of the ordered product.
Target	Shows the target amount.

1.Data Import & Transformation

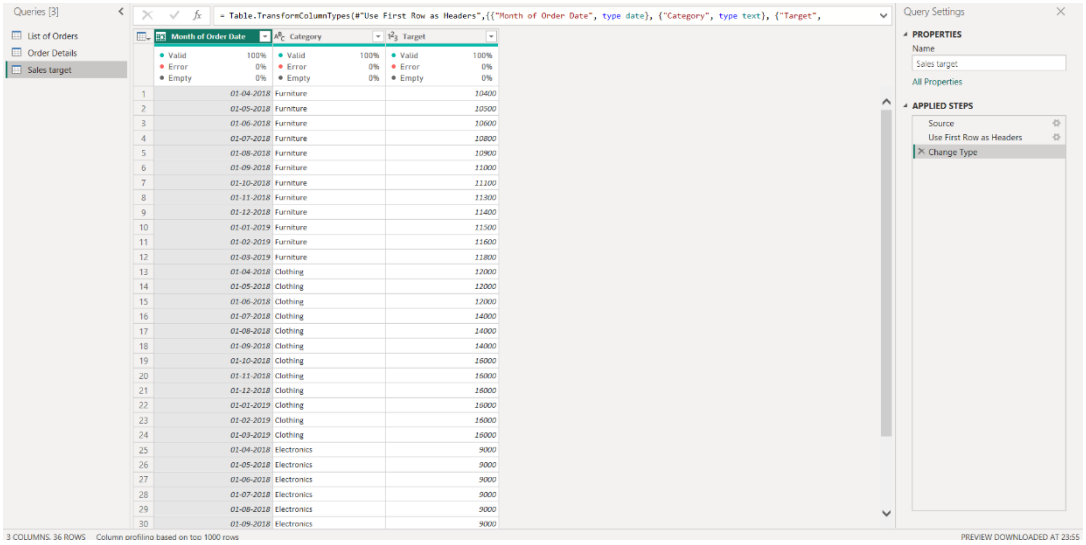
- ❖ Import List of Orders.csv into Power BI.



- ❖ Open the list of orders in the Power Query Editor using the Transform Data option

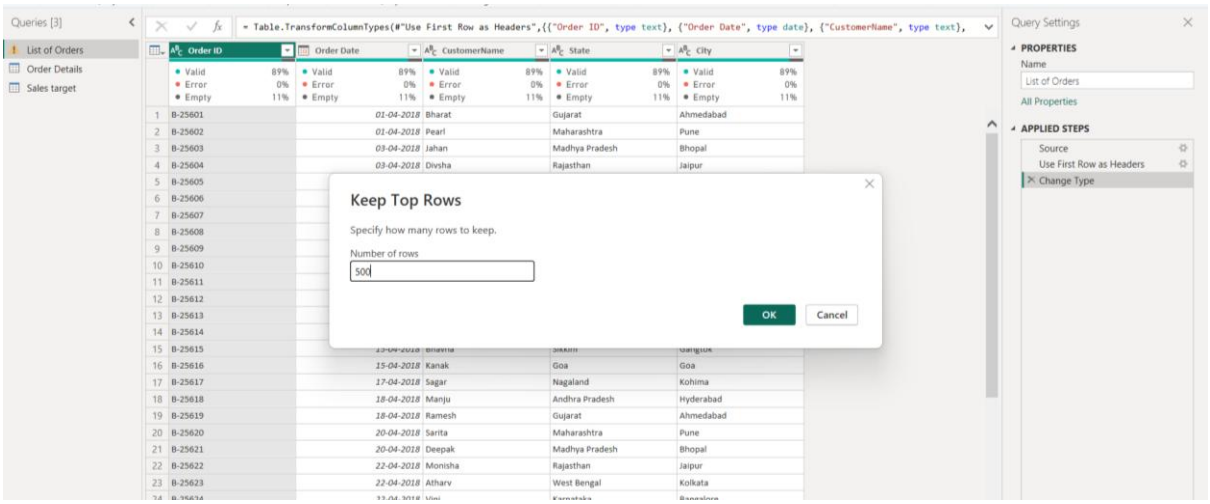


❖ Import Order Details.csv and Sales Target.csv into Power Query Editor.

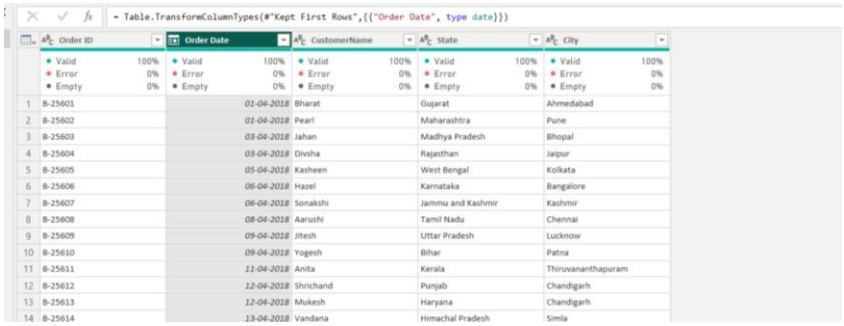


2. Row Limiting & Data Types:

❖ Restrict the List of Orders table to the first 500 rows



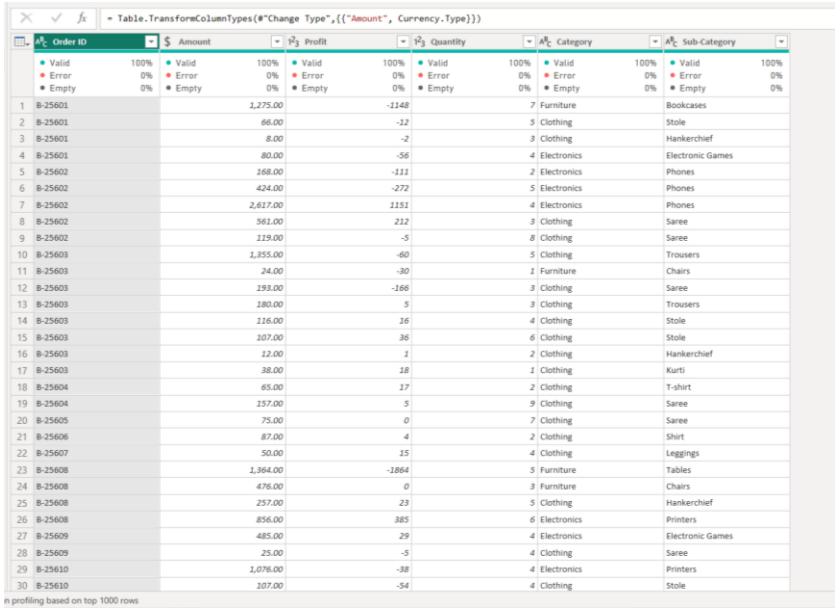
❖ Set the Order Date column to the Date data type.



```
= Table.TransformColumnTypes(#"Kept First Rows",{{"Order Date", type date}})
```

	Order ID	Order Date	CustomerName	State	City
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla

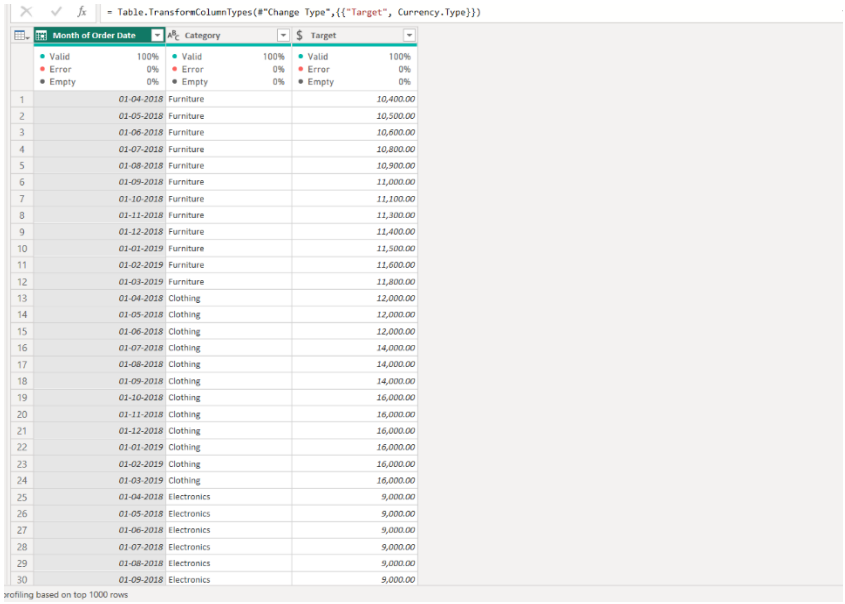
❖ Change the Amount and Target columns to fixed decimal numbers.



```
= Table.TransformColumnTypes(#"Change Type",{{"Amount", Currency.Type}})
```

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankerchief
4	B-25601	80.00	-56	4	Electronics	Electronic Games
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankerchief
17	B-25603	38.00	18	1	Clothing	Kurti
18	B-25604	65.00	17	2	Clothing	T-shirt
19	B-25604	157.00	5	9	Clothing	Saree
20	B-25605	75.00	0	7	Clothing	Saree
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	50.00	15	4	Clothing	Leggings
23	B-25608	1,364.00	-1864	5	Furniture	Tables
24	B-25608	476.00	0	3	Furniture	Chairs
25	B-25608	257.00	23	5	Clothing	Hankerchief
26	B-25608	856.00	385	6	Electronics	Printers
27	B-25609	485.00	29	4	Electronics	Electronic Games
28	B-25609	25.00	-5	4	Clothing	Saree
29	B-25610	1,076.00	-38	4	Electronics	Printers
30	B-25610	107.00	-54	4	Clothing	Stole

n profiling based on top 1000 rows



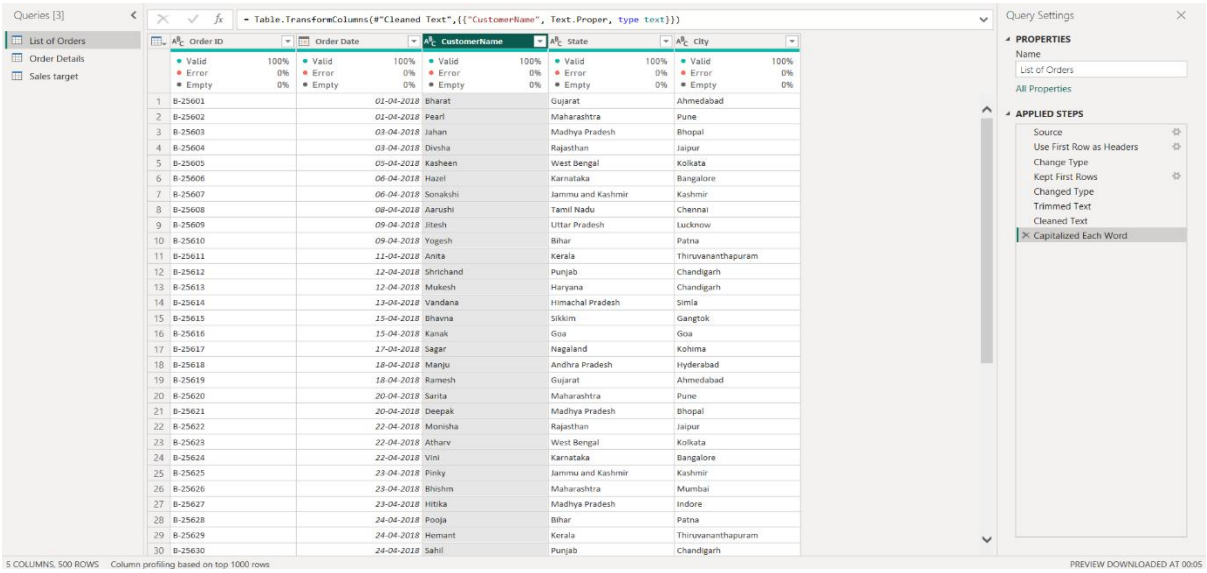
```
= Table.TransformColumnTypes(#"Change Type",{{"Target", Currency.Type}})
```

	Month of Order Date	Category	Target
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	01-04-2018	Furniture	10,400.00
2	01-05-2018	Furniture	10,500.00
3	01-06-2018	Furniture	10,600.00
4	01-07-2018	Furniture	10,800.00
5	01-08-2018	Furniture	10,900.00
6	01-09-2018	Furniture	11,000.00
7	01-10-2018	Furniture	11,100.00
8	01-11-2018	Furniture	11,300.00
9	01-12-2018	Furniture	11,400.00
10	01-01-2019	Furniture	11,500.00
11	01-02-2019	Furniture	11,600.00
12	01-03-2019	Furniture	11,800.00
13	01-04-2018	Clothing	12,000.00
14	01-05-2018	Clothing	12,000.00
15	01-06-2018	Clothing	12,000.00
16	01-07-2018	Clothing	14,000.00
17	01-08-2018	Clothing	14,000.00
18	01-09-2018	Clothing	14,000.00
19	01-10-2018	Clothing	16,000.00
20	01-11-2018	Clothing	16,000.00
21	01-12-2018	Clothing	16,000.00
22	01-01-2019	Clothing	16,000.00
23	01-02-2019	Clothing	16,000.00
24	01-03-2019	Clothing	16,000.00
25	01-04-2018	Electronics	9,000.00
26	01-05-2018	Electronics	9,000.00
27	01-06-2018	Electronics	9,000.00
28	01-07-2018	Electronics	9,000.00
29	01-08-2018	Electronics	9,000.00
30	01-09-2018	Electronics	9,000.00

rofiling based on top 1000 rows

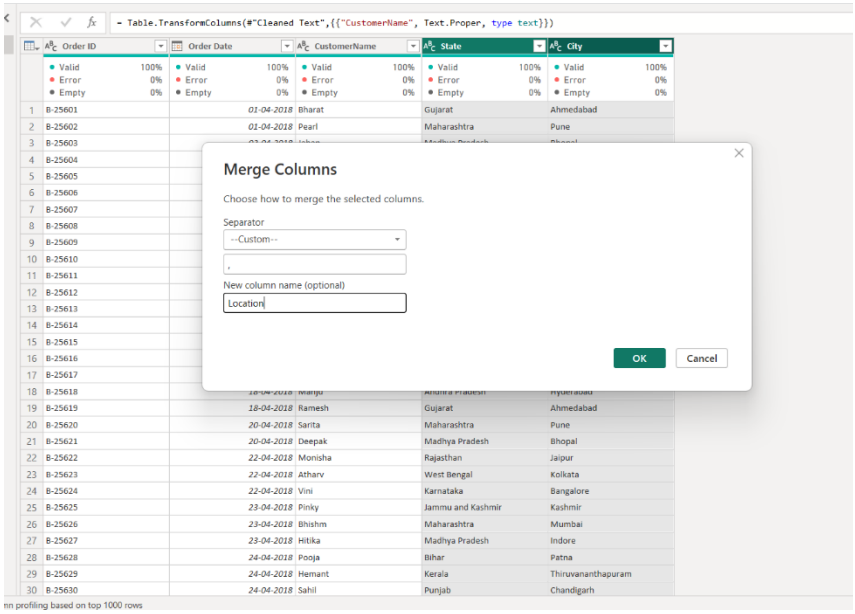
3. Text Formatting:

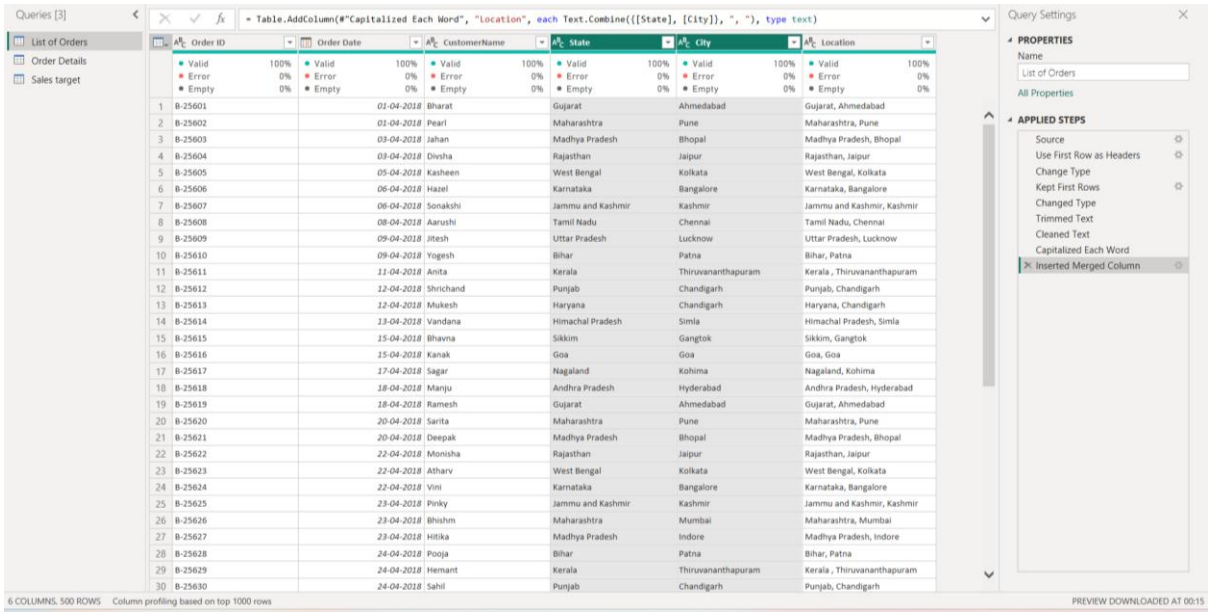
- ❖ Format the Customer Name column to proper case to ensure consistent capitalization.



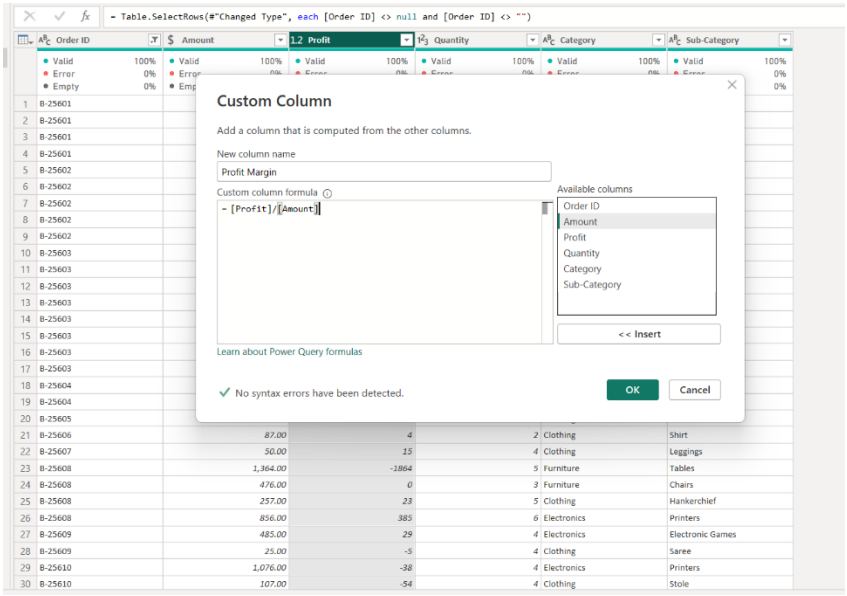
4. Column Creation:

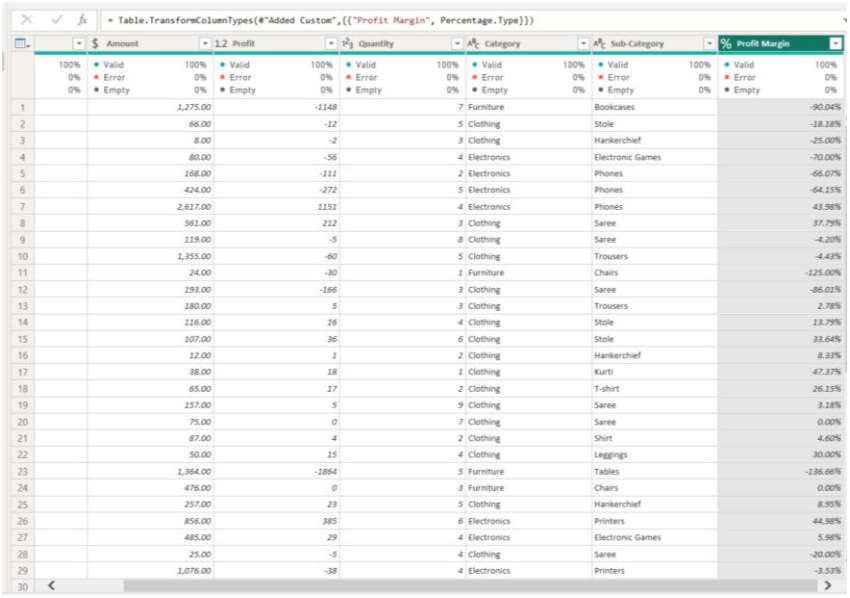
- ❖ Merge the City and State columns to create a new column named Location in the format: City, State





❖ Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

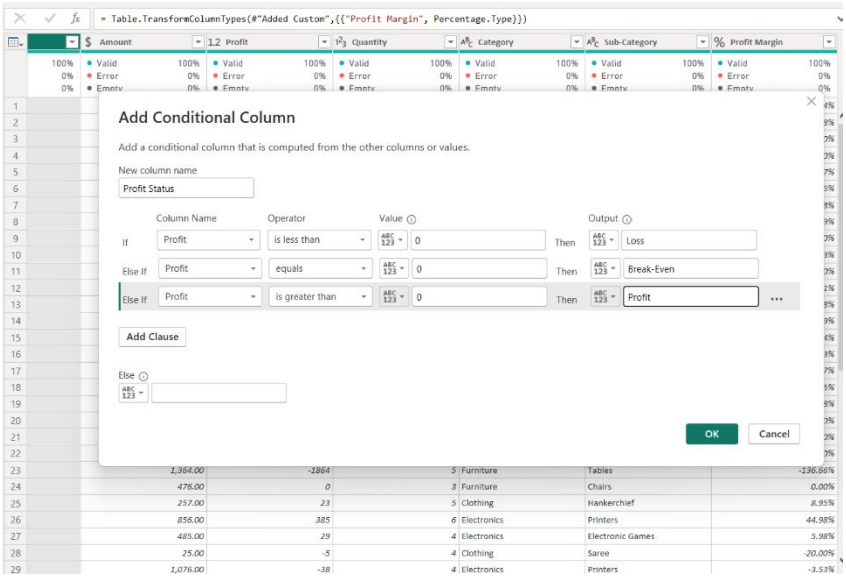




	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
1	1,275.00	-1148	7	Furniture	Bookcases	-90.04%
2	66.00	-12	5	Clothing	Stole	-18.18%
3	8.00	-2	3	Clothing	Hankerchief	-25.00%
4	80.00	-56	4	Electronics	Electronic Games	-70.00%
5	168.00	-111	2	Electronics	Phones	-66.07%
6	424.00	-272	5	Electronics	Phones	-64.15%
7	2,617.00	1151	4	Electronics	Phones	43.98%
8	561.00	212	3	Clothing	Saree	37.79%
9	119.00	-5	8	Clothing	Saree	-4.20%
10	1,355.00	-60	5	Clothing	Trousers	-4.43%
11	24.00	-30	1	Furniture	Chairs	-125.00%
12	193.00	-166	3	Clothing	Saree	-86.01%
13	180.00	5	3	Clothing	Trousers	2.78%
14	116.00	16	4	Clothing	Stole	13.79%
15	107.00	36	6	Clothing	Stole	33.64%
16	12.00	1	2	Clothing	Hankerchief	8.33%
17	38.00	18	1	Clothing	Kurti	47.37%
18	65.00	17	2	Clothing	T-shirt	26.15%
19	157.00	5	9	Clothing	Saree	3.18%
20	75.00	0	7	Clothing	Saree	0.00%
21	87.00	4	2	Clothing	Shirt	4.60%
22	50.00	15	4	Clothing	Leggings	30.00%
23	1,364.00	-1864	5	Furniture	Tables	-136.66%
24	476.00	0	3	Furniture	Chairs	0.00%
25	257.00	23	5	Clothing	Hankerchief	8.95%
26	856.00	385	6	Electronics	Printers	44.98%
27	485.00	29	4	Electronics	Electronic Games	5.98%
28	25.00	-5	4	Clothing	Saree	-20.00%
29	1,076.00	-38	4	Electronics	Printers	-3.53%

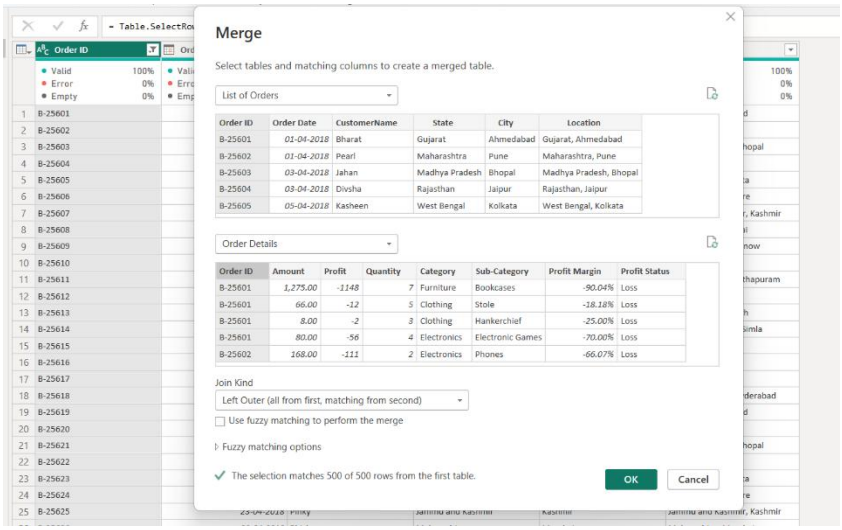
5. Conditional Column:

- ❖ Add a conditional column named Profit Status based on:
  - Profit < 0 → Loss
  - Profit = 0 → Break-Even
  - Profit > 0 → Profit



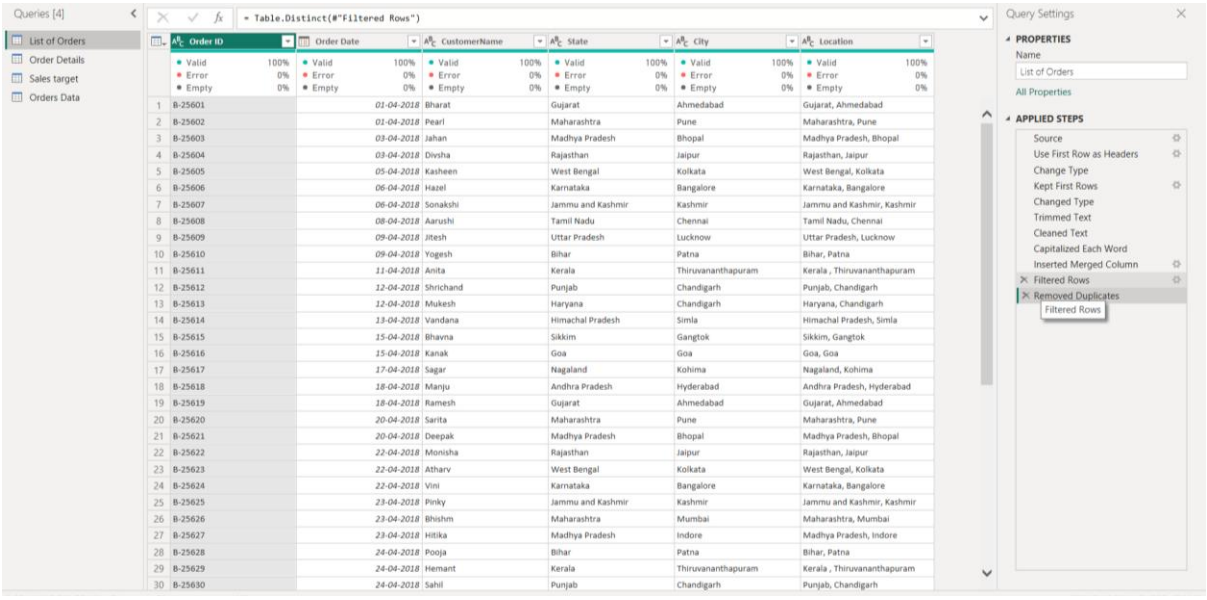
6. Merging Data (Joins)

- ❖ Merge the List of Orders and Order Details tables using Order ID.
- ❖ Name the merged output table as Orders Data.

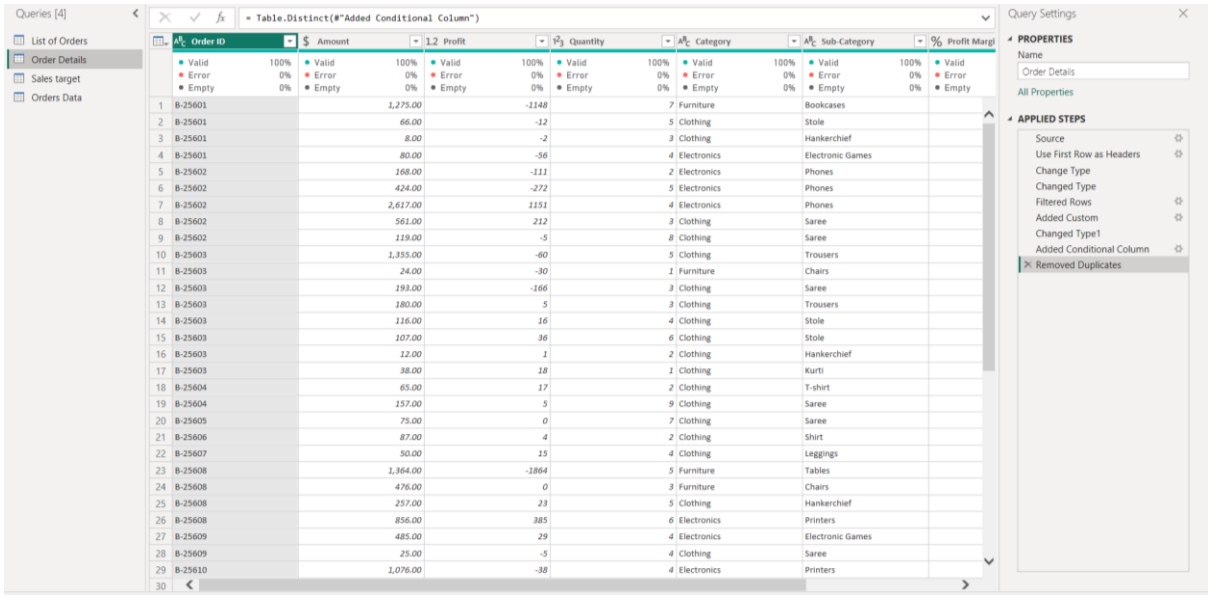


7. Handling Missing Data & Duplicate Data

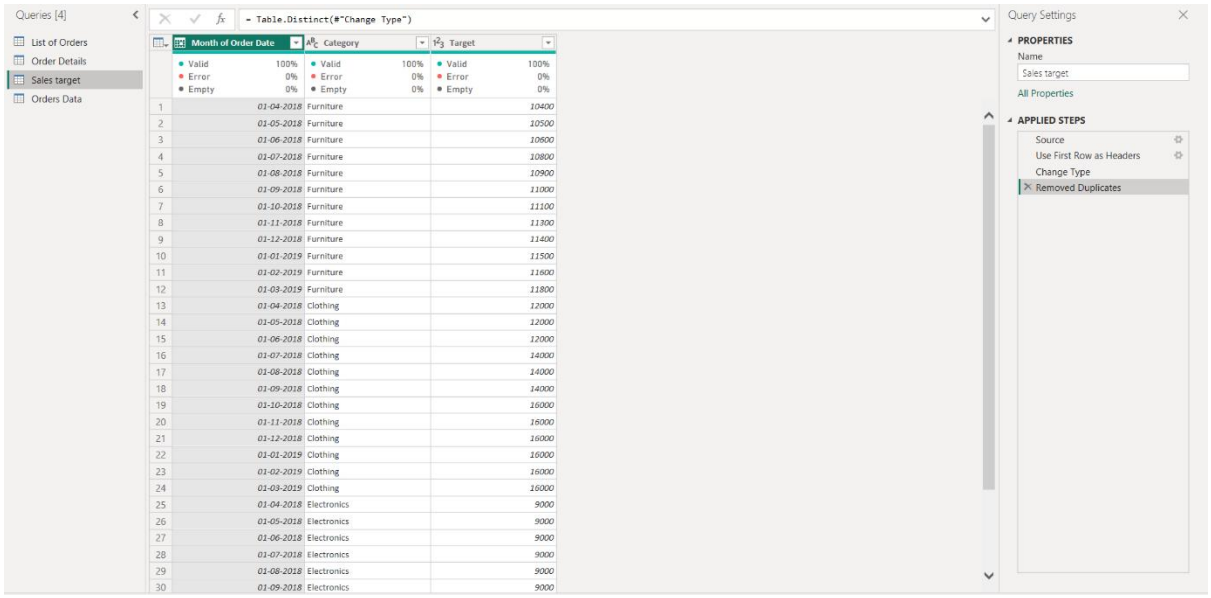
- ❖ Removing Duplicate in List of Order table using Order ID



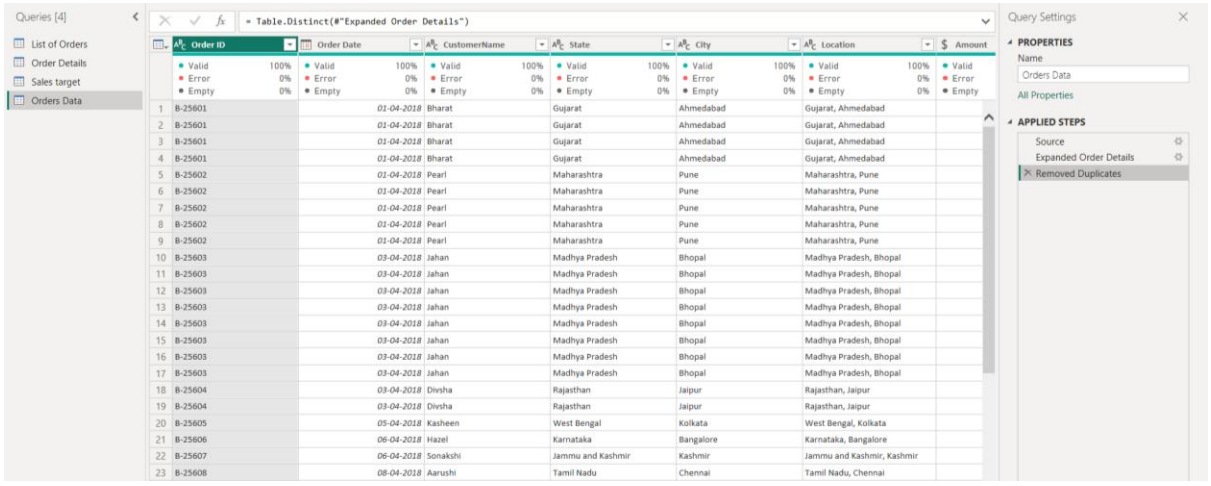
❖ Removing Duplicate in Order Details, Sales target and order data table using Order ID



	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankerchief
4	B-25601	80.00	-56	4	Electronics	Electronic Games
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankerchief
17	B-25603	38.00	18	1	Clothing	Kurti
18	B-25604	65.00	17	2	Clothing	T-shirt
19	B-25604	157.00	5	9	Clothing	Saree
20	B-25605	75.00	0	7	Clothing	Saree
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	50.00	15	4	Clothing	Leggings
23	B-25608	1,364.00	-1864	5	Furniture	Tables
24	B-25608	476.00	0	3	Furniture	Chairs
25	B-25608	257.00	23	5	Clothing	Hankerchief
26	B-25608	856.00	385	6	Electronics	Printers
27	B-25609	485.00	29	4	Electronics	Electronic Games
28	B-25609	25.00	-5	4	Clothing	Saree
29	B-25610	1,076.00	-38	4	Electronics	Printers



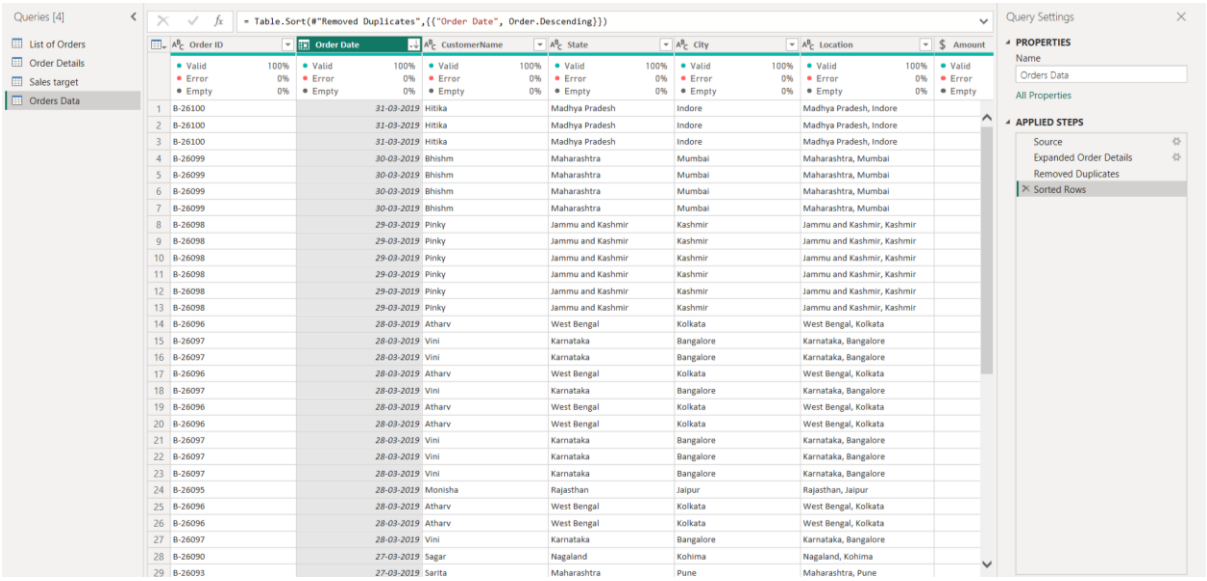
	Month of Order Date	Category	Target
1	01-04-2018	Furniture	10400
2	01-05-2018	Furniture	10500
3	01-06-2018	Furniture	10600
4	01-07-2018	Furniture	10800
5	01-08-2018	Furniture	10900
6	01-09-2018	Furniture	11000
7	01-10-2018	Furniture	11100
8	01-11-2018	Furniture	11300
9	01-12-2018	Furniture	11400
10	01-01-2019	Furniture	11500
11	01-02-2019	Furniture	11600
12	01-03-2019	Furniture	11800
13	01-04-2018	Clothing	12000
14	01-05-2018	Clothing	12000
15	01-06-2018	Clothing	12000
16	01-07-2018	Clothing	14000
17	01-08-2018	Clothing	14000
18	01-09-2018	Clothing	14000
19	01-10-2018	Clothing	16000
20	01-11-2018	Clothing	16000
21	01-12-2018	Clothing	16000
22	01-01-2019	Clothing	16000
23	01-02-2019	Clothing	16000
24	01-03-2019	Clothing	16000
25	01-04-2018	Electronics	9000
26	01-05-2018	Electronics	9000
27	01-06-2018	Electronics	9000
28	01-07-2018	Electronics	9000
29	01-08-2018	Electronics	9000
30	01-09-2018	Electronics	9000



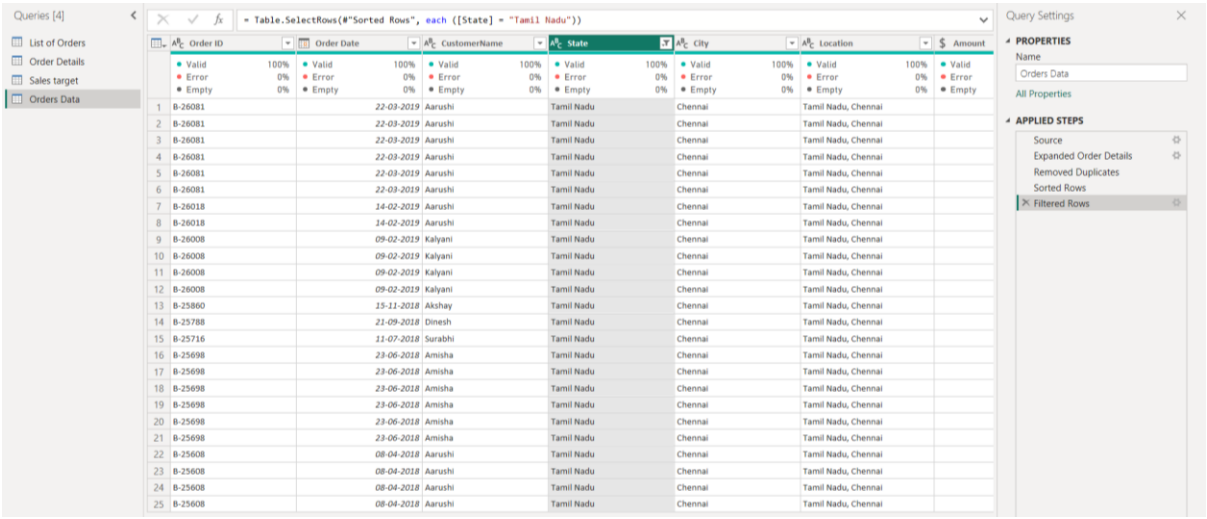
	Order ID	Order Date	CustomerName	State	City	Location	Amount
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat, Ahmedabad	
2	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat, Ahmedabad	
3	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat, Ahmedabad	
4	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Gujarat, Ahmedabad	
5	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra, Pune	
6	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra, Pune	
7	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra, Pune	
8	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra, Pune	
9	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Maharashtra, Pune	
10	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh, Bhopal	
11	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh, Bhopal	
12	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh, Bhopal	
13	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh, Bhopal	
14	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh, Bhopal	
15	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh, Bhopal	
16	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh, Bhopal	
17	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Madhya Pradesh, Bhopal	
18	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Rajasthan, Jaipur	
19	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Rajasthan, Jaipur	
20	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	West Bengal, Kolkata	
21	B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Karnataka, Bangalore	
22	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Jammu and Kashmir, Kashmir	
23	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Tamil Nadu, Chennai	

8. Sorting and Filtering Data

- ❖ Sort records in the Orders Data table by Order Date in descending order.



- ❖ Apply filters to analyse orders from a specific state (e.g., Tamil Nadu).



9. Grouping and Aggregating Data

- ❖ Duplicate the Order Details table and perform the following:
  - ✓ Count of Order ID



  - ✓ Average Profit by Category



Category	Average profit
Furniture	20.44
Electronics	-19.38
Clothing	15.29

  - ✓ Total Amount by Sub-Category



Sub-Category	Total amount
Bookcases	₹ 25,800
Phones	₹ 11,691
Trousers	₹ 7,109
T-shirt	₹ 3,205
Saree	₹ 17,831
Shirt	₹ 2,535
Leggings	₹ 851
Tables	₹ 3,967
Electronic Games	₹ 11,942
Printers	₹ 16,822
Chairs	₹ 5,724
Hankerchief	₹ 4,362
Stole	₹ 7,583
Skirt	₹ 822
Furnishings	₹ 4,295
Accessories	₹ 9,472
Kurti	₹ 1,154

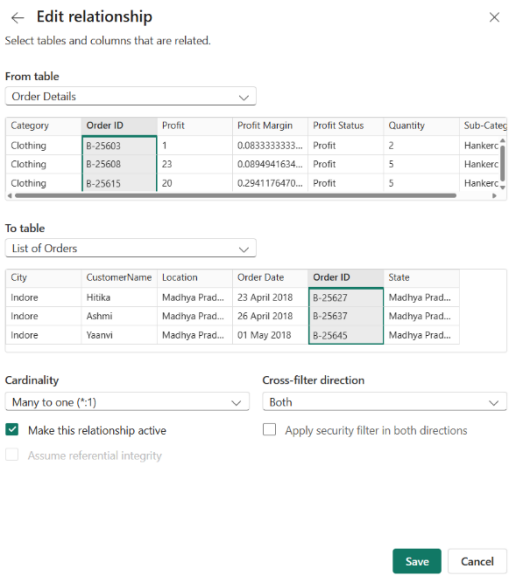
- ❖ Duplicate the Sales Target table and aggregate the total Target by Month of order date.



Month	Total target
January	43500
February	43600
March	43800
April	31400
May	31500
June	31600
July	33800
August	33900
September	34000
October	36100
November	36300
December	36400

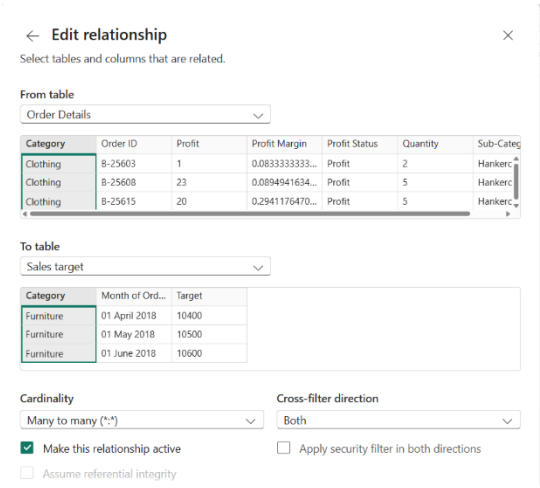
10. Data Modelling:

- Establish a relationship between:
  - ❖ List of Orders and Order Details using Order ID

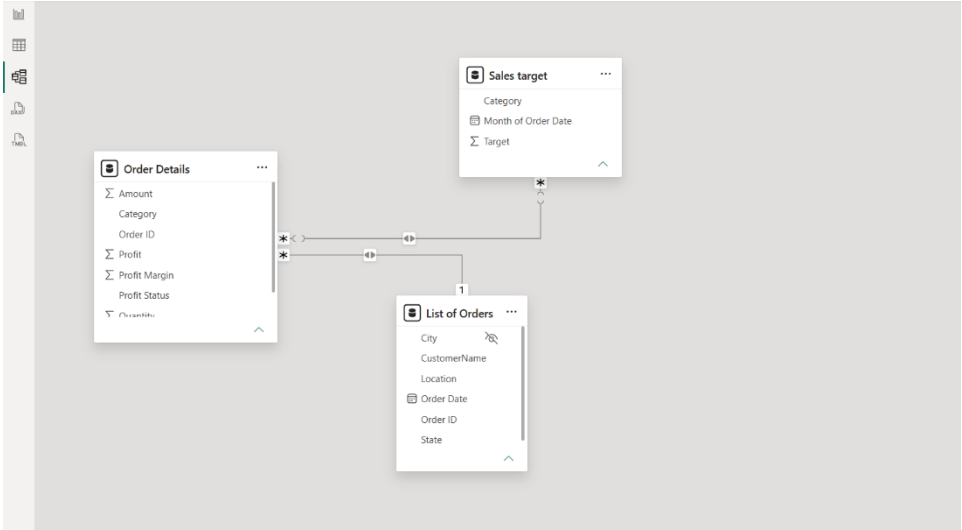


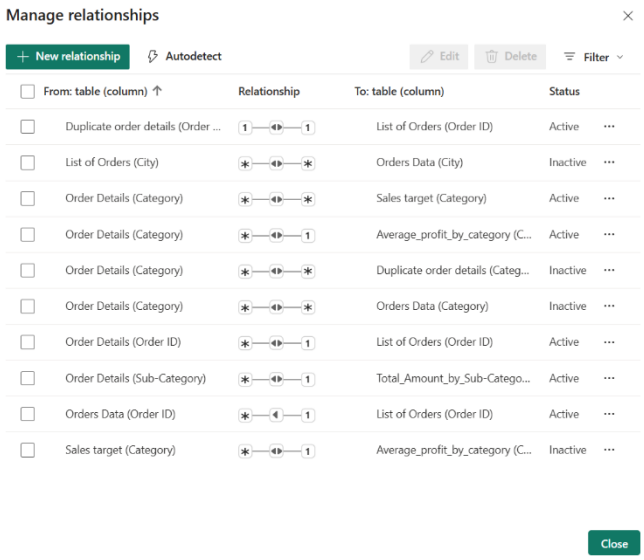
Establish a relationship between:

- ❖ Order Details and Sales Target using Category



❖ Use Manage Relationships to ensure relationships are active and correctly defined.





✅ Conclusion

- This project successfully demonstrates the use of Power BI for data transformation, cleaning, and modelling of e commerce datasets.
- The three datasets, List of Orders, Order Details, and Sales Target, were processed and organized to improve data consistency and analytical usability.
- Key transformations, table merging, data aggregation, and relationship modelling established a structured foundation for analysing order distribution, category wise profitability, sub category sales performance, and monthly sales targets.
- Overall, the project highlights the importance of data preparation in converting raw data into meaningful insights that support effective business analysis and informed decision making.